

电机规格承认书

Motor Spec Approval Sheet

客户编号 (Customer No.) : _____

电机型号 (Motor Model) : YD4025S-002

管理编号 (Management number) : YD-25-03S-005

版本号 (Version number) : A

承认原因 (Admit That Reason)	☒ 新产品承认 NEW APPROVE ☐ 规格变更再承认 CHANGE CODE APPROVE AGAIN
客户确认签名 (Customer Approved)	

发放日期 Issuance Date	电机设计 Design of motor	工程批准 Engineering Approved BY
2025-03-13	刘文果	JOHN

注释 REMARKS:

1. 本产品规格书供客户确认:

除非本产品规格书另行修订, 此技术指标将作为贵司产品来料检验标准;

在接受本技术规格并经确认之后, 请在订购产品之前回签给我们;

如果贵方 (未封样品) 未回签给我方, 我方即认本产品规格书已被贵方认可。

This product specification is for customer's confirmation;

Unless this product specification is otherwise revised, this technical index will be used as the incoming inspection standard of your products.

After acceptance and confirmation of this product specification, please sign back to us before ordering the product;

If you (unsealed sample) do not sign back to us, we acknowledge that this specification has been approved by you.

2. 当出现问题时, 双方以本产品规格书确认标准进行商议。

In case of any problem, both parties shall confirm the standard of this product specification for negotiation.

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1. 适用范围及对象/Scope and object of application:

本规格书适用于电机型号 YD4025S-002，应用于家用电器、个人护理、医疗、运动器材、办公自动化等。

This specification sheet is applicable to motor model YD4025S-002, which is used in household appliances, personal care, medical care, sports equipment, office automation, etc.

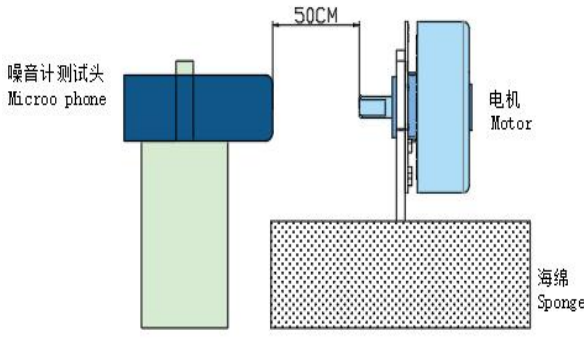
2. 标准工作条件/Standard operating conditions:

序号 No.	项目 Item	规格 Specification
2-1	安装方向 Direction of mountin	输出轴方向水平 Horizontal to output shaft direction
2-2	电源 Power source	D. C regulated power source, which has no doubt about the results, shall be used. 确认测量时无疑问的可调节直流稳压电源
2-3	环境 Temperature, Humidity	除非另有规定, 标准范围的大气状况进行测量和测试如下 环境温度 : -10℃ to 50℃ 相对湿度 : 25% to 80 % 空气压力 : 86kPa to 106 kPa Unless otherwise specified, the standard range of atmospheric conditions for making measurements and tests are as follows: Ambient temperature : -10℃ to 50℃ Relative humidity : 25% to 80 % Air pressure : 86kPa to 106 kPa

3. 结构特征/CONSTRUCTION

序 号 No.	项目 Item	规格 Specification
3-1	定子槽数 Number of stator slots	12 Slots
3-2	相数•极数	3 相 14 极 3 Phases • 14 Poles
3-3	驱动电路 Drive circuit	3 相 无霍尔元件 3-phase Houle element
3-4	接线方式 Circuit Type	星型接法 / Y Connection
3-5	相间电阻 Resistance between phases	0.36 Ω ± 10%
3-6	结构 Configuration	开放型. 外转子 / Open type Outer rotor
3-7	马达重量: Weight	约 160 克/台 about 160g/unit
3-8	外观 Appearance	不含有机机械损坏及不充分的浸蚀等 Does not contain mechanical damage and insufficient erosion.
3-9	外观尺寸 External dimension	参考外形图 Reference contour drawing

4. 电机特性/Motor characteristics:

序号 No.	项目 Item	规格 Specification	
4-1	额定电压 Ratedvoltage	12V DC	
4-2	使用电压范围 Range of operating voltage	10~14.8 V DC	
4-3	空载电流 (A) No load current(A)	0.6A Max	
4-4	空载转速 (rpm) No load speed (rpm)	3600RPM±10%	
4-5	负载电流 (A)Rated load current (A)	2.0A±15%	
4-6	额定转速 (rpm) Rated load speed (rpm)	3000RPM±10%	
4-7	额定力矩 Rated torque	60mN.m	
4-8	回转方向 Rotation Direction	CW/CCW	从输出轴方向看 From the direction of the output axis
4-9	线圈绝缘等级 Coil insulation class	B 种 Class B	
4-10	绝缘电阻 Insulation Resistance	DC 500V, 10[M Ω] Min	DC500(V) 马达安装部位, 线圈间 DC500(V) Motor mounting position, between coils
4-11	绝缘耐压 Insulation withstand voltage	500V AC 1(s) or 1 mA Max	在电机端子和机壳之间加 AC500V 1 秒钟, 遮 断电流: 1mA Max Apply AC500V for 1 second between the motor terminals and the case, with a maximum current of 1mA.
4-12	噪音 Noise	55dB(A) Max DC[12V], 空载运行, 背景噪音 32 dBA 30cm 距离测试 DC[12V], no-load operation, background noise 32 dBA, 50cm distance test	
		 噪音计测试头 Microo phone 50CM 电机 Motor 海绵 Sponge	

4-13	寿命时间 Life time	测试循环, 8000 小时最小 Test cycle, 8000 hours minimum 无负载额定电压 常温 $20 \pm 5 [^{\circ}\text{C}]$ No load Rated voltage Normal temperature $20 \pm 5 [^{\circ}\text{C}]$	判定基准 (参考标准) /Determination Criteria (Reference Criteria) : 1) 额定转速/Rated speed: 不低于初始值-30%; 不高于初始值+30% data-30% Initial Min data+60% Initial Max 2) 额定电流/Rated current: 不大于初始值± 30%/Initialata±30% max
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5. 特性测定条件及方法/TEST METHOD:

1. 通常在温度为 25°C ，相对湿度为 65% 的条件测试，在温度为 $5^{\circ}\text{C} \sim 35^{\circ}\text{C}$ 和相对湿度为 40%~85% 时也视为正常的测试环境。

Temperature 25°C , Relative humidity 65% as standard. However a range of $5^{\circ}\text{C} \sim 35^{\circ}\text{C}$ and humidity of 25%~80% can be viewed as normal test conditions.

2. 测试方法 (Test Method) :

(1). 额定电流、转速、功率，须在额定电压下，运行 1 分钟取数。

Observe rated current . rated speed and rated input power after operating for 1 min.

(2). 噪音.Noise

无负荷(Duty 100%).

Test method No Load

环境噪音:Background noise:32 dBA Reference.

把马达放在厚 30mm 泡沫（海棉）板上，马达轴芯水平位置放置，噪音计距马达外周 50cm.

Placed the motor on a sponge cushion of thickness 30mm and the shaft should be positioned horizontally , the microphone position is 50cm from motor.

6. 包装/PACKING:

根据适当的尺寸标准在一个标准纸箱中装 60~200 个产品。

One standard carton contains 60~200 pcs products according to appropriate size criterion.

7. 使用时注意事项/When using the matters needing attention:

7-1. 由于是外转子结构，稳固性能很差，超过 6cm 摔落即可能损坏转子，要注意轻拿轻放， 严禁堆叠；

Because it is an external substructure, the stability performance is very poor, more than 6cm fall may damage the rotor, should pay attention to gently handle,

Do not stack;

7-2. 请不要在电气接通时锁住电机轴，造成异常停止。

Do not lock the motor shaft while the electric is on.

7-3. 因为电压过高会损坏马达，请不要在本规格书规定的电压范围外使用马达。

Since the high voltage will damage the micro motor, do not use the voltage over the rated.

7-4. 电机接通电后，不要经常性堵住机轴，电机轴端压入零件时，必须顶着轴的另一端，防止电机内部移动，影响电机正常使用。

After the motor is connected, do not often block the shaft. When the shaft end of the motor presses into the parts, it must be pressed against the other end of the shaft to prevent the internal movement of the motor, which affects the normal use of the motor.

7-5. 禁止给电机直接冲击。

Do not give direct shock to the motor.

7-6. 在本产品跌落后或者受到较大的冲击后，因可能产生电机机身受损，导致电机运转不良、声音不良等问题，请不要使用本产品（即使初期运转正常，我们也将不保证产品的性能）。

Please do not use this product (even if the initial operation is normal, we will not guarantee the performance of the product) because the motor fuselage may be damaged, which may lead to poor motor operation, bad sound and other problems.

7-7. 不允许在高温高湿或灰尘环境中工作，请不要在产生有害气体及有害物质场所使用；（如硫、低分子硅氧烷、赛安体系、福尔马林体系、快干胶或 H₂S、SO₂、Cl 等腐蚀性气体存在的环境）。另外，如成品机器内存有上述物质，请事前充分进度行确认，可能会造成电机一定程度生锈、缩短电机寿命。

Do not allow to work in high temperature, high humidity or dust environment, please do not use in the production of harmful gas and harmful substances; (such as sulfur, low molecular siloxane, Seyan system, formalin system, fast drying glue or H₂S, SO₂, Cl and other corrosive gases exist). In addition, if the finished machine memory has the above substances, please fully confirm the progress in advance, may cause a certain degree of motor rust, shorten the life of the motor.

7-8. 库存尽量避免在上述有害气体环境、高温高湿环境、化学腐蚀中存储产品，请在 3 个月内使用。当电机在高温或低温使用时，由于温度突然急剧变化，表面上可能会产生水滴附着在电机上，请给予足够的时间让电机在室温下储存。

Inventory Avoid storing products in the above harmful gas environment, high temperature and humidity environment, and chemical corrosion. Please use the products within 3 months. When the motor is used at high or low temperature, water droplets may be produced on the surface to adhere to the motor due to sudden and sharp changes in temperature. Please allow sufficient time for the motor to be stored at room temperature.

7-9. 有电子零件会受到周向带电物体的感应电场和人体电荷的影响，电机操作人员和工作台必须接地以避免 静电破坏。另外，生产装置确保没有电流泄漏。

Connect an operator and working table to the ground when handle to avoid the electrostatic destruction of the electronic parts due to the induction of

circumferential charged objects, electric field and the charge of human body, additionally, the production installments should have no current leakage

- 7-10.** 用于本产品的所有材料必须符合 RoHS 及 REACH 等环保要求, 而且必须是有记载的且为安全的物质, 不可使用特殊的化学物质; 但是如果以后 RoHS 指令发生变更而出现了问题的情况下, 则需另行协商。

All materials used in this product must meet environmental requirements such as RoHS and REACH, and must be documented and safe Substances, do not use special chemical substances; However, if problems arise due to changes in the RoHS instructions, they need to be negotiated separately.

- 7-11.** 在有金属片、金属粉、水蒸气、水滴、以及可能会造成短路的环境: 如大量灰尘、粉尘、水蒸气、水滴等侵入到马达内部的环境下使用, 则不在保证范围内。

In the presence of sheet metal, metal powder, water vapor, water droplets, and environments that may cause short circuits, such as large amounts of dust, dust, Water vapor, water droplets and other intrusions into the motor internal environment under the use of, is not guaranteed.

- 7-12.** 噪音、浪涌、电压跌落、静电、电路破坏以及原子引线干扰等测试, 请在贵司的整机状态下进行, 特别是带有长导线的部品, 请确认干扰等项目。

Noise, surge, voltage drop, static electricity, circuit damage and atomic lead interference should be tested on your machine State. Especially for parts with long wires, please confirm the interference items.

- 7-13.** 注意处理马达的工作环境, 避免含铁质或磁性等物体被马达吸住, 否则马达将会产生噪声, 性能会波动, 从而减少产品可靠性。

Pay attention to the working environment of the motor to avoid objects containing iron or magnetism being absorbed by the motor, otherwise the motor will produce noise and performance fluctuation, so as to reduce the reliability of the product.

- 7-14.** 电气噪声、机械噪声、振动、旋转寿命和电共振等电机和电路故障, 只有在电机组装好后才能确认, 用户应该使用前测试确认。

For item such as electric noise, mechanical noise, vibration, fluctuation in rotation life drift and electric resonance between motor and circuit which are only accessible after the motor is assembled in a set users should per-test such characteristics in the real set

- 7-15.** 与成品装配时请留意装配位置是否正确, 如不当会产生噪音、振动, 以致影响轴承寿命、产生声音。

Although the motor monomer meets the performance requirements, but also due to the impact of the real machine characteristics change, so please make a judgment after the final assembly with the real machine.

- 7-16.** 电机故障事项发生时根据本规格书的记载事项, 双方协议后进行解决。

When defective performance is occurred mutual discussions shall be made in accordance with specifications to resolve the matter.

7-17. 如果超过本规格书规定值，则不能保证电机正常的性能。

Proper performance do not be guaranteed if the used in excess of specified value than indicated in the specifications

7-18. 本规格书内容的变更或追加，需要双方互相讨论和协商同意后进行。

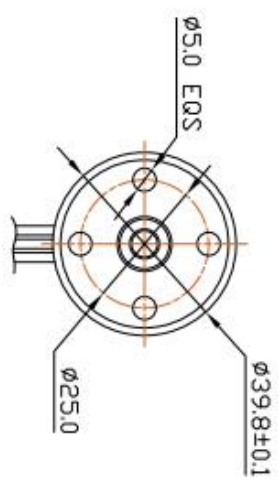
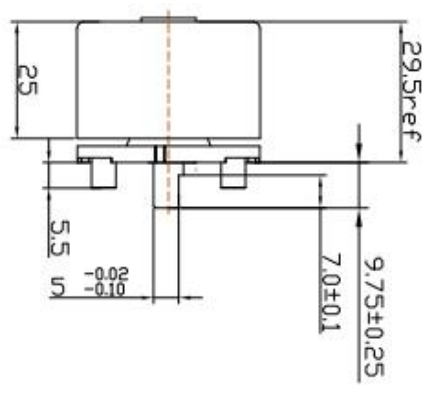
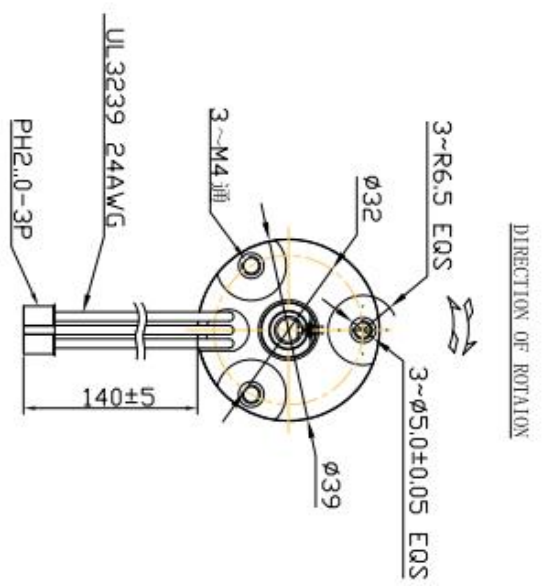
Any revision and addition to the specification shall be made after mutual discussion and agreement

8-19. 本规格书有中文·英文版。如有哪方出现差异的情况下、优先以中文为准。

This specification is available in Chinese and English. In case of any discrepancy between the two parties, the Chinese version shall prevail.

7-20. 如果对本产品的安全性有任何疑问，请务必联系敝司，敝司会进行相关的技术验证。

If you have any doubt about the safety of this product, please be sure to contact our company, our company will carry out the relevant technical verification.



Remark:

1. Rated voltage range 10~14.8V, rated voltage 12V.
2. Motor axial end clearance: none.
3. When one end of the motor shaft is pressed into the parts, the other end of the shaft must be clamped to prevent the internal movement of the motor, thus affecting the normal use of the motor;
4. Materials meet RoHS requirements.

				未注尺寸公差	设计	刘文果	2025-03-13	产品名称	YD4025S-002
				0	±0.15	绘图		零件编号	YD4-MT0-4025002
				.0	±0.10	审核		客户代码	/
				.00	±0.05	批准	2025-03-13	样办单号	20250313008
				0°	±1°	比例	1 : 1		
序号	位置	变更内容	签名	日期	特殊公差参照国标	视图	1 : 1	亿达电机 VIDAMOTOR 东莞市亿达电机科技有限公司 YIDA MOTOR TECHNOLOGY Co., Ltd.	Rev. 版本 A